

# Gour Gupal Talukder Shawon

Shahjalal University of Science and Technology, Sylhet, Bangladesh  
Sylhet, Bangladesh | [gourgupaltalukder@gmail.com](mailto:gourgupaltalukder@gmail.com) | +880 1746-244930  
[Portfolio](#) | [Google Scholar](#) | [LinkedIn](#) | [github.com/gfor-gour](https://github.com/gfor-gour)

## Education

**Shahjalal University of Science and Technology**  
*B.Sc. in Software Engineering* CGPA: 3.52/4.00

*Sylhet, Bangladesh*  
Mar 2022 - May 2026

## Research Experience

### HiveMind Research Group

*Research Contributor*

*Remote*

Jan 2026 - Present

**Paper:** *AgentCollabBench: Diagnosing When Good Agents Make Bad Collaborators*

*FAGEN, ICML 2026*

**Resources:** [arXiv Preprint](#) | [GitHub](#) | [HuggingFace Dataset](#)

*Under Review (NeurIPS 2026)*

- Contributed to *AgentCollabBench*, a benchmark for evaluating collaboration failures and reliability in multi-agent LLM systems.
- Assisted with dataset design, benchmark construction, and evaluation setup for Software Engineering, DevOps, and Data Engineering tasks.
- Participated in experiment design, dataset validation, and research discussions within a distributed AI research team.

**Paper:** *Relevance is Not Obligation: Action-Relative Memory Activation for Tool-Using Agents*

**Status:** *Abstract Submitted (ICLR 2027)*

- Contributed to the formulation of a dynamic obligation-aware framework that evaluates memory-action pairs at the action boundary to enforce hard constraints and mitigate agent execution failures
- Spearheaded dataset annotation and quality control for coding-domain memory-action pairs, validating action scopes, footprint parameters, and enforcement tiers under near-miss interference.

### Undergraduate Thesis

May 2025 – Nov 2025

*Commit Message Generation using LLMs with Issue-Based Context*

*Under Review (Q4)*

- **Co-First & Corresponding Author:** Led experimental design, benchmark curation, human evaluation pipeline, and paper manuscript.
- Evaluated GPT-4 Turbo, Claude 3.5, and Gemini 1.5 Pro on 309 issue-linked commits from real-world open-source repositories. [\[HuggingFace Dataset\]](#)
- Developed a dual evaluation framework combining automated NLP metrics and double-blind assessment by 30 software developers.
- Improved commit message quality by 42–53% over human-written baselines through prompt engineering.

### Advanced Machine Intelligence Research Lab, AIUB

*Remote*

*Intern Research Assistant*

Feb 2026 - Jun 2026

- Contributing to research on federated unlearning for LoRA-tuned LLMs, focusing on privacy-preserving machine learning and machine unlearning.
- Assisting with literature review, experimental design, and evaluation for federated learning and parameter-efficient fine-tuning methods.
- Supporting an ongoing computer vision project on diffusion-based restoration of snow-degraded images.

## Competitive Programming

- Solved **1100+** programming problems across [LeetCode](#), [Vjudge](#), and Codeforces with a focus on algorithms, data structures, and problem-solving.
- Maintained a **433-day** LeetCode streak, earned **23** badges, and achieved a peak rating of **1622**.

## Industry Experience

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### ShellBeeHaken Ltd.

Intern Software Engineer (Onsite)

Dhaka, Bangladesh

Dec 2025 - May 2026

- Contributed to **production UI components** and navigation flows using **React Native (Expo)** and **Next.js**, integrating **REST APIs** and managing application state across screens.
- Assisted in building **event-driven backend services** with **AWS Lambda**; gained hands-on exposure to distributed systems concepts including **data replication** and **load balancing** during architecture discussions.
- Participated in the full **API development cycle** from writing data contracts to **integration testing**, while working in an **Agile** team with sprint planning and code reviews.

## Projects

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### Personality Detection Using Machine Learning

[GitHub](#) | [Live](#)

- Improved model **F1 score by 20%** via advanced **feature engineering** and **hyperparameter tuning**, achieving **91% accuracy** in real-time personality prediction.
- Tools: Python, scikit-learn, Flask, Next.js, Pandas, NumPy, joblib.

### Local Household Service Provider Platform

[GitHub](#) | [Live](#)

- Designed an **optimized data-fetching architecture** that reduced retrieval time by **60%**, resulting in **20% higher** overall application responsiveness and user satisfaction.
- Integrated **Google Gemini-powered AI chatbot**, improving service query response rate by **60%**; authored critical system documents including SRS and project proposal.
- Tools: Next.js, Express.js, MongoDB, Nodemailer, Cloudinary.

### Real-Time Multi-Module Chat Application

[GitHub](#)

- Built a **real-time desktop chat system** using **Java, Socket.IO (Netty), PostgreSQL, and Swing**, structured with a **multi-module Maven architecture** and shared DTOs for consistent client-server communication.
- Implemented **WebSocket-based messaging** with a custom **reconnection strategy (exponential backoff)** and heartbeat monitoring to ensure resilient real-time connectivity.
- Designed an **event-driven backend architecture** supporting authentication, messaging, and file transfer workflows, with secure and efficient persistence using **HikariCP, PostgreSQL indexing, and BCrypt authentication**.

## Technical Skills

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**Languages** JavaScript, Java, C++, Python, TypeScript

**Frontend** Next.js, React, React Native, Flutter

**Backend** Node.js, Express.js, RESTful APIs, GraphQL, Socket.IO

**Databases** MongoDB, Firebase, MySQL, PostgreSQL

**Tools** AWS, Docker, Git, Linux, Serverless (AWS Lambda), Postman, Maven

## Academic Honors & Certifications

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- **2nd Runner-up** - "Bridging Engineering & Computer Science" International Hackathon, organized by ILM-AI, co-hosted by **St. George's, University of London**. Built a browser-based projectile motion simulator with optional air resistance (two-member team). [GitHub](#) | [Live](#)
- **Certifications:** AWS Cloud Practitioner · AWS Lambda · AWS DynamoDB · AWS API Gateway · Claude Code in Action (Anthropic) · Generative AI (Udacity & AWS) · ML Crash Course (Google).